Mathematics practice booklet: Grade 11 expectations

Grade 11 · Practice booklet

Name: _____

What this booklet is

This is general practice written for Mathematics at Grade 11 level, following the Common Core State Standards for Mathematics standards for that grade. It is not made from your child's MAP questions, and it does not replace the classroom. Use it as a guide to what a child is expected to know and do at this level.

How to use it

- Do one section at a time, about 10 minutes, 3 days a week. Stop while it is still enjoyable.
- Read each question all the way through, twice, before answering. Underline the key words and say what is being asked.
- Take your time. A careful, correct answer is worth more than a fast one.
- Use scrap paper and show the working. Then check the answer before moving on.
- Ask your child to explain how they got an answer, even when it is right.
- If a section feels too hard, do the section before it again, or ask the teacher. That is part of learning, not a problem.

Draft: to be checked by a teacher before use.

Section 1: Patterns, equations and algebra thinking

Standards practised: A-SSE.B.3, A-APR.B.2, A-APR.B.3, A-APR.D.6, A-REI.A.2, F-BF.A.1, F-BF.B.3, F-BF.B.4, F-LE.A.2, F-LE.A.4

A worked example

Solve 3^x = 20. Give the answer to 2 decimal places.

1. Take the logarithm of both sides: log(3^x) = log(20).
2. Bring the power down: x log(3) = log(20).
3. Divide: x = log(20) / log(3).
4. Use a calculator: x = 1.3010 / 0.4771, which is about 2.73.
5. Check: 3^2.73 is about 20.1, close to 20.

x is about 2.73

Questions

1. Factor: x^2 - 5x - 14

2. The graph of y = x^2 is moved to give g(x) = (x - 3)^2 + 4. Describe the move.

3. Find the remainder when P(x) = x^3 - 2x^2 + x - 6 is divided by x - 2.

4. Show that x = 1 is a zero of x^3 - 6x^2 + 11x - 6, then find the other zeros.

5. Simplify, and say which values of x are not allowed: (x^2 - 9) / (x^2 + 3x)

6. Solve: 2 / (x - 1) = 3 / (x + 2)

7. Solve and check: the square root of (x + 7) = x + 1

8. A nursery in Ras Al Khaimah plants 400 mangrove seedlings and the number grows by 5% each year. Write a function for the number after t years and find the number after 10 years.

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9. Evaluate: log base 2 of 32, and log base 10 of 0.001. Then solve log base 3 of x = 4.

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10. Solve 4^x = 30 using logarithms. Give the answer to 2 decimal places.

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11. Let f(x) = x + 2 and g(x) = x^2. Find f(g(3)) and g(f(3)). Are they the same?

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12. Find the inverse of f(x) = 2x + 3. Then show that f(f^-1(5)) = 5, and explain what the inverse does.

Section 2: Shapes, measuring and space

Standards practised: G-SRT.C.6, G-SRT.C.7, G-SRT.C.8, G-SRT.D.11, F-TF.A.1, F-TF.A.2, F-TF.B.5, F-TF.C.8

A worked example

From a point 100 m from the base of a tower in Abu Dhabi, the angle of elevation of the top is 40 degrees. Find the height of the tower to the nearest metre.

1. Draw a right triangle: the side next to the angle is 100 m and the side opposite is the height h.
2. Opposite and adjacent give the tangent: tan 40 degrees = h / 100.
3. h = 100 x tan 40 degrees = 100 x 0.8391.
4. h is about 83.9 m, which is 84 m to the nearest metre.

About 84 m

Questions

1. A right triangle has sides 5, 12 and 13 (13 is the longest). For the angle A opposite the side 5, find sin A, cos A and tan A. Then find cos B for the other acute angle B and compare it with sin A.

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2. A 6 m ladder leans against a wall and makes an angle of 70 degrees with the ground. How high up the wall does it reach? Give the answer to 2 decimal places.

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3. A ramp rises 1.2 m over a horizontal distance of 10 m. Find the angle of the ramp with the ground, to 1 decimal place.

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4. Convert 150 degrees to radians, and 3 pi / 4 radians to degrees.

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5. A Ferris wheel has radius 20 m. How far does a seat travel along the circle as the wheel turns through pi / 3 radians? Give the answer to 1 decimal place.

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6. Give the exact values of sin 30 degrees, cos 45 degrees and tan 60 degrees.

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7. On the unit circle, give the coordinates of the points at angles 90 degrees, 180 degrees and 120 degrees.

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8. Find the amplitude and the period of y = 2 sin(3x).

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9. The angle theta is in the second quadrant and sin theta = 3/5. Find cos theta and tan theta.

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10. In triangle ABC, angle A = 40 degrees, angle B = 60 degrees and side a = 8 cm (opposite A). Use the law of sines to find side b, to 1 decimal place.

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11. Two sides of a triangular garden are 7 m and 9 m, and the angle between them is 50 degrees. Use the law of cosines to find the third side, to 1 decimal place.

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12. Salem says that sin 2x is always twice sin x. Test x = 30 degrees and decide whether he is right.

Section 3: Data, graphs and chance

Standards practised: S-ID.A.2, S-IC.A.1, S-IC.B.3, S-IC.B.4, S-IC.B.5, S-ID.A.4

A worked example

The marks on a school test are roughly normal with mean 72 and standard deviation 8. About what percentage of students scored between 64 and 88?

1. 64 is one standard deviation below the mean (72 - 8) and 88 is two above (72 + 16).
2. In a normal distribution about 34% lies from the mean to 1 standard deviation, and about 47.5% from the mean to 2 standard deviations.
3. Add the two parts: 34% + 47.5% = 81.5%.

About 81.5%

Questions

1. Find the mean and the standard deviation of the data 4, 6, 8, 10, 12. Give the standard deviation to 2 decimal places.

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2. A school in Sharjah has 1,200 students. The principal asks 50 of them about the canteen. What is the population and what is the sample?

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3. Hamdan wants to know how many hours the students in his school sleep. He asks only the students who play in the school football team. Is this a good sample? Suggest a better way.

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4. In a random sample of 120 students, 36 walk to school. Estimate how many of the 1,500 students in the school walk.

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5. A poll of a random sample finds that 62% prefer a new park design, with a margin of error of 4%. Between which percentages is the true value of the whole population likely to lie?

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6. Poll A asks 100 people and poll B asks 1,000 people, both chosen at random. Which usually has the smaller margin of error, and why?

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7. A study gives 30 plants a new fertiliser and 30 similar plants none. The plants are assigned to the two groups at random. Is this an experiment or an observational study? Why is the random assignment important?

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8. After 4 weeks the fertilised plants have a mean height of 14.2 cm and the others 11.8 cm. What is the difference? What else would you want to know before deciding the fertiliser works?

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9. Test scores are roughly normal with mean 70 and standard deviation 10. About what percentage of scores lie between 60 and 80?

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10. For the same scores, about what percentage are above 90?

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11. Using the same distribution (mean 70, standard deviation 10), find the z-score of 85. Layla has a mark with z-score -2: what is her mark?

Section 4: Numbers and calculating

Standards practised: N-CN.A.1, N-CN.A.2, N-CN.C.7, N-RN.A.1, N-RN.A.2, F-BF.A.2, A-SSE.B.4

A worked example

Write (2 + 3i)(1 - i) in the form a + bi.

1. Multiply each part: 2 x 1 + 2 x (-i) + 3i x 1 + 3i x (-i).
2. That is 2 - 2i + 3i - 3i^2.
3. Since i^2 = -1, the term -3i^2 equals +3.
4. Collect: (2 + 3) + (-2i + 3i) = 5 + i.

5 + i

Questions

1. Simplify the square root of -49 using i.

2. Add: (3 + 2i) + (5 - 4i)

3. Multiply: (4 + i)(2 - 3i)

4. Simplify i^7.

5. Write the complex conjugate of 4 - 3i and find the product of the number and its conjugate.

6. Divide and write in the form a + bi: (6 + 2i) / (1 + i)

7. Solve x^2 + 2x + 5 = 0.

8. Evaluate 8^(2/3) and 16^(3/4).

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9. Write the cube root of x^5 using a rational exponent, and simplify x^(1/2) times x^(3/2).

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10. An arithmetic sequence begins 5, 9, 13, 17. Write a rule for the nth term and find the 20th term.

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11. A geometric sequence begins 3, 6, 12, 24. Find the common ratio and the 8th term.

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12. Maryam saves AED 5 on day 1 and doubles the amount she saves each day for 7 days (5, 10, 20, and so on). How much has she saved in total? Why does this grow faster than saving AED 5 more each day?

Answer guide for parents

Section 1: Patterns, equations and algebra thinking

1. (x - 7)(x + 2) (numbers that multiply to -14 and add to -5)
2. 3 units right and 4 units up. The vertex moves from (0, 0) to (3, 4).
3. -4 (the remainder is P(2) = 8 - 8 + 2 - 6)
4. 1 - 6 + 11 - 6 = 0, so x - 1 is a factor. Dividing gives x^2 - 5x + 6 = (x - 2)(x - 3). The zeros are 1, 2 and 3.
5. (x - 3) / x, with x not equal to 0 or -3. (Factor to (x - 3)(x + 3) / (x(x + 3)) and cancel x + 3.)
6. x = 7 (cross multiply: 2x + 4 = 3x - 3). Check: 2/6 = 3/9, both are 1/3.
7. x = 2. Squaring gives x + 7 = x^2 + 2x + 1, so x^2 + x - 6 = 0 and x = 2 or x = -3. Checking: x = -3 gives 2 = -2, which is false, so it is rejected.
8. N = 400 x 1.05^t. After 10 years, N is about 652 (400 x 1.6289).
9. 5, -3, and x = 81. (2^5 = 32; 10^-3 = 0.001; 3^4 = 81.)
10. x = log(30) / log(4), which is about 2.45. (Check: 4^2 = 16 and 4^3 = 64, so x lies between 2 and 3.)
11. f(g(3)) = f(9) = 11 and g(f(3)) = g(5) = 25. They are different, so the order of composition matters.
12. f^-1(x) = (x - 3) / 2. f^-1(5) = 1 and f(1) = 5. The inverse undoes what f does, swapping inputs and outputs.

Section 2: Shapes, measuring and space

1. sin A = 5/13, cos A = 12/13, tan A = 5/12. cos B = 5/13, the same as sin A, because the two acute angles add up to 90 degrees.
2. 6 x sin 70 degrees, which is about 5.64 m
3. About 6.8 degrees (the angle whose tangent is 1.2 / 10 = 0.12)
4. 150 degrees = 5 pi / 6 radians. 3 pi / 4 radians = 135 degrees. (180 degrees = pi radians.)
5. Arc length = radius x angle = 20 x pi / 3, which is about 20.9 m
6. 1/2, the square root of 2 divided by 2, and the square root of 3
7. (0, 1), (-1, 0) and (-1/2, the square root of 3 divided by 2). The point at angle t is (cos t, sin t).
8. Amplitude 2; period 2 pi / 3. (The period of sin(bx) is 2 pi divided by b.)
9. cos theta = -4/5 and tan theta = -3/4. (sin^2 + cos^2 = 1 gives cos^2 = 16/25, and cosine is negative in the second quadrant.)
10. b = 8 x sin 60 / sin 40, which is about 10.8 cm
11. c^2 = 49 + 81 - 2 x 7 x 9 x cos 50, which is about 49.0, so c is about 7.0 m
12. No. sin 60 degrees is about 0.866, but 2 x sin 30 degrees = 2 x 0.5 = 1. One example that does not match is enough to show the rule is false.

Section 3: Data, graphs and chance

1. Mean 8. The squared distances from the mean are 16, 4, 0, 4, 16, which add to 40; 40 / 5 = 8 and the square root of 8 is about 2.83.

2. The population is all 1,200 students. The sample is the 50 students who were asked.
3. No, it is biased: athletes may not sleep like everyone else. Better: pick students at random, for example by drawing random numbers from the class register.
4. 36 / 120 = 30%, and 30% of 1,500 is about 450 students.
5. Between 58% and 66% (62% minus 4% and 62% plus 4%).
6. Poll B. A larger random sample gives results that vary less from one sample to another, so the estimate is more precise.
7. It is an experiment. Random assignment makes the two groups alike on average, so a difference in growth is more likely to be caused by the fertiliser.
8. The difference is 2.4 cm. You would want to know how much heights vary within each group (the spread) and whether the difference is bigger than the variation expected by chance, for example by repeating the experiment.
9. About 68% (within 1 standard deviation of the mean)
10. About 2.5%. The score 90 is 2 standard deviations above the mean; 95% lie within 2 standard deviations, leaving 5% in the two tails, so 2.5% in the upper tail.
11. z = (85 - 70) / 10 = 1.5. Layla's mark is 70 - 2 x 10 = 50.

Section 4: Numbers and calculating

1. 7i (because i^2 = -1)
2. 8 - 2i (add the real parts and the imaginary parts separately)
3. 11 - 10i (8 - 12i + 2i - 3i^2 = 8 - 10i + 3)
4. -i (i^4 = 1, so i^7 = i^3 = -i)
5. The conjugate is 4 + 3i. The product is 16 + 9 = 25, a real number.
6. 4 - 2i. (Multiply top and bottom by 1 - i: the top is 8 - 4i and the bottom is 2.)
7. x = -1 + 2i or x = -1 - 2i (the discriminant is 4 - 20 = -16, and the square root of -16 is 4i)
8. 4 and 8. (8^(2/3): the cube root of 8 is 2, then square it. 16^(3/4): the fourth root of 16 is 2, then cube it.)
9. x^(5/3), and x^2 (add the exponents: 1/2 + 3/2 = 2)
10. a_n = 5 + 4(n - 1), so the 20th term is 5 + 4 x 19 = 81.
11. The ratio is 2, and the 8th term is 3 x 2^7 = 384.
12. 5 x (2^7 - 1) / (2 - 1) = AED 635. Doubling is exponential growth (multiplying by the same number each day), which overtakes adding a fixed amount; adding AED 5 more each day would give only 5 + 10 + ... + 35 = AED 140.